PREDICT SOIL BEARING CAPACITY USING MACHINE LEARNING MODEL

A Project Report

Submitted in partial fulfilment of the Requirements for the award of the degree of

Bachelor of Technology

In

Civil Engineering with Computer Applications

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DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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ABSTRACT

Soil bearing capacity is a fundamental parameter in geotechnical engineering that determines the load-carrying capacity of soil for supporting structural foundations. Traditional methods for estimating bearing capacity rely on laboratory testing, empirical correlations, and theoretical equations such as those developed by Terzaghi, Meyerhof, and Hansen. These methods are often time-consuming, expensive, and limited by assumptions regarding soil homogeneity and ideal conditions.

Recent advances in **Machine Learning (ML)** and **data-driven modelling** provide powerful alternatives for predicting geotechnical properties using historical datasets. ML algorithms can capture complex nonlinear relationships between soil properties and bearing capacity, enabling faster and more accurate predictions. **This project investigates the application of machine learning models to predict soil bearing capacity using soil index properties and geotechnical parameters of Muzaffarpur region.** Several algorithms including **Random Forest, Support Vector Machine, and Artificial Neural Networks** are implemented using Python. Model performance is evaluated using metrics such as **R^2, RMSE, and MAE.** **R^2 value is 0.98 (98 %) which indicates higher accuracy of trained data with tested data.**

The project highlights the integration of **civil engineering domain knowledge with computer applications**, demonstrating the potential of ML for improving geotechnical design efficiency.

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CHAPTER 1

INTRODUCTION

1.1 General

Soil plays a very important role in civil engineering because every structure such as buildings, bridges, dams, roads, and towers is supported by soil. The strength and stability of any structure depend largely on the ability of the soil to support the loads transmitted through the foundation. Therefore, understanding soil behaviour and determining its load-carrying capacity is essential before constructing any engineering structure.

One of the most important parameters in geotechnical engineering is the **bearing capacity of soil**. Bearing capacity refers to the maximum load per unit area that soil can safely support without causing failure or excessive settlement. If the soil does not have sufficient bearing capacity, the structure may experience settlement, cracks, tilting, or even complete collapse. For this reason, engineers must carefully evaluate soil properties before designing foundations.

Traditionally, the bearing capacity of soil is calculated using theoretical formulas developed by geotechnical engineers. These formulas are based on soil mechanics principles and consider factors such as soil cohesion, internal friction angle, unit weight of soil, and depth of foundation. One of the most widely used theories is **Terzaghi's Bearing Capacity Theory**, which provides an equation to calculate the ultimate bearing capacity of shallow foundations.

However, traditional analytical methods sometimes require complex calculations and assumptions. In many real-world situations, soil conditions are highly variable and difficult to represent using simple equations. With the advancement of computer technology, engineers are now using **data-driven techniques such as machine learning** to analyze soil behavior and predict engineering parameters more efficiently.

Machine learning techniques allow computers to learn patterns from data and make predictions without explicitly programming mathematical equations. By using large datasets of soil parameters, machine learning models can identify relationships between different variables and produce accurate predictions. These techniques have recently gained attention in the field of civil engineering for solving complex problems related to soil mechanics, structural analysis, and construction management.

The present project focuses on developing a **machine learning-based system for predicting soil bearing capacity**. The system uses soil parameters as inputs and predicts both ultimate and safe bearing capacity values. The aim of this project is to integrate traditional geotechnical engineering concepts with modern computer applications to create a fast, reliable, and user-friendly prediction system.

1.2 Soil Bearing Capacity

Bearing capacity is defined as the maximum pressure that soil can safely support without undergoing shear failure. When a foundation is constructed, the load of the structure is transferred to the soil through the foundation. If the applied load exceeds the soil's bearing capacity, the soil may fail or undergo excessive settlement.

There are three types of bearing capacity commonly used in foundation design:

1. Ultimate Bearing Capacity (qu)

Ultimate bearing capacity is the maximum load per unit area that soil can support before failure occurs. It represents the theoretical maximum load that soil can withstand.

2. Safe Bearing Capacity (qs)

Safe bearing capacity is the allowable load per unit area that soil can safely carry without failure. It is obtained by dividing the ultimate bearing capacity by a factor of safety.

3. Allowable Bearing Capacity

Allowable bearing capacity considers both shear failure and settlement limitations to ensure safe foundation performance.

Several factors influence soil bearing capacity, including:

- Soil type
- Soil cohesion
- Angle of internal friction
- Unit weight of soil
- Depth of foundation

Engineers determine these parameters through laboratory tests and field investigations such as **Standard Penetration Test (SPT)**, **plate load test**, and soil sampling.

1.3 Types of Foundations

A foundation is the lowest part of a structure that transfers loads from the structure to the soil. Foundations are generally classified into two main categories.

1. Shallow Foundations

Shallow foundations are used when the soil near the ground surface has sufficient bearing capacity. These foundations are constructed at relatively shallow depths.

Common types include:

- **Strip footing** – Used for load-bearing walls.
- **Isolated footing** – Used for individual columns.
- **Combined footing** – Supports two or more columns.
- **Raft foundation** – Supports multiple columns over a large area.

2. Deep Foundations

Deep foundations are used when the soil near the surface is weak and cannot support structural loads. These foundations transfer loads to deeper, stronger soil layers.

Common types include:

- **Pile foundation** • **Pier foundation** • **Caisson foundation**

Selecting the appropriate foundation type depends on soil conditions, load requirements, and construction feasibility.

1.4 Importance of Soil Bearing Capacity

Determining soil bearing capacity is essential for safe and economical construction. Accurate estimation of bearing capacity helps engineers design foundations that can safely support structural loads.

Some important benefits of determining soil bearing capacity include:

- Ensures structural stability and safety
- Prevents excessive settlement of structures
- Reduces the risk of foundation failure
- Helps in selecting the appropriate foundation type
- Improves overall structural performance
- Reduces construction costs by avoiding overdesign

If the bearing capacity of soil is underestimated, the foundation may become unnecessarily large and expensive. On the other hand, overestimating the bearing capacity may lead to structural failure. Therefore, accurate prediction of soil bearing capacity is very important in civil engineering practice.

1.5 Introduction to Machine Learning

Machine learning is a branch of **artificial intelligence (AI)** that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed. Instead of using fixed mathematical formulas, machine learning models analyze data to identify relationships between input variables and output results.

Machine learning techniques are widely used in various fields such as:

- Healthcare
- Finance
- Transportation
- Manufacturing
- Engineering

In machine learning, data is usually divided into two categories:

1. Input Features

These are independent variables used to predict results. In this project, soil parameters such as cohesion, friction angle, and foundation depth are input features. **2. Target Variables**

These are output values predicted by the model. In this project, the outputs are ultimate bearing capacity and safe bearing capacity.

Machine learning algorithms such as **Random Forest, Support Vector Machines, Artificial Neural Networks, and Decision Trees** are commonly used for prediction tasks.

1.6 Role of Machine Learning in Civil Engineering

In recent years, machine learning has been widely applied in civil engineering to solve complex problems. Many engineering problems involve large datasets and nonlinear relationships between variables. Machine learning models can effectively analyze such data and produce accurate predictions.

Applications of machine learning in civil engineering include:

- Prediction of soil properties
- Structural health monitoring
- Traffic flow prediction
- Construction cost estimation
- Earthquake damage assessment
- Material strength prediction

In geotechnical engineering, machine learning can help predict soil properties such as compressibility, shear strength, settlement, and bearing capacity. These predictions can assist engineers in making quick and reliable decisions during foundation design.

1.7 Scope of the Project

The scope of this project is to develop a **machine learning-based system for predicting soil bearing capacity** using soil parameters and foundation properties.

The project involves the following tasks:

- Generation of a soil dataset using Terzaghi's bearing capacity equation.
- Preprocessing and encoding of soil parameters.
- Training a machine learning model using the Random Forest algorithm.
- Predicting ultimate and safe bearing capacity values.
- Developing a web application using the Flask framework for user interaction.
- Storing prediction results in a SQLite database.

The developed system allows users to input soil parameters and obtain bearing capacity predictions quickly. This approach demonstrates how machine learning can be integrated with civil engineering concepts to improve efficiency and decision-making in foundation design.

CHAPTER 2

OBJECTIVE

The main objectives of the project “Predicting Soil Bearing Capacity Using Machine Learning Models” are as follows:

1. To apply machine learning techniques for predicting the ultimate and safe bearing capacity of soil based on input parameters.
2. To develop a machine learning model using Random Forest Regression, which can learn patterns from the dataset and provide accurate predictions.
3. To develop a web-based application interface using the Flask framework that allows users to easily input soil parameters and obtain bearing capacity predictions.
4. To store prediction results in a SQLite database for record keeping and future analysis.

CHAPTER 3

LITERATURE REVIEW

The prediction of soil bearing capacity has been an important topic in geotechnical engineering for many years. Accurate estimation of bearing capacity is necessary for the safe design of foundations. Over time, several researchers have developed theoretical methods, experimental studies, and computational models to determine the load carrying capacity of soil. In recent years, modern techniques such as machine learning have also been used to improve the prediction accuracy of soil properties. This chapter presents a review of the important studies related to soil bearing capacity and the application of machine learning in civil engineering.

3.1 Historical Development of Bearing Capacity Theory

The study of soil bearing capacity began in the early development of soil mechanics. One of the most important contributions in this field was made by **Karl Terzaghi**, who is known as the father of soil mechanics. In 1943, Terzaghi proposed a theoretical equation to determine the ultimate bearing capacity of shallow foundations. His equation considered soil parameters such as cohesion (c), angle of internal friction (φ), unit weight of soil (γ), and foundation dimensions. Terzaghi also introduced the bearing capacity factors **N_c, N_q, and N_γ**, which depend on the angle of internal friction of the soil.

Terzaghi's equation is widely used for analyzing shallow foundations such as strip footings, square footings, and circular footings. The theory assumes that soil failure occurs due to shear failure beneath the foundation. Although Terzaghi's theory provides a good approximation of soil behaviour, it is based on several assumptions and may not always represent real soil conditions accurately

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Later, many researchers modified Terzaghi's theory to improve its accuracy and applicability. **Meyerhof (1951)** extended the bearing capacity equation by introducing correction factors such as shape factor, depth factor, and load inclination factor. These modifications made the theory more suitable for different types of foundations and loading conditions.

Another significant contribution was made by **Hansen (1970)**, who proposed a generalized bearing capacity equation that included additional factors for ground surface inclination and load inclination. Hansen's method provided more flexibility in analyzing complex foundation problems.

Vesic (1973) further improved the bearing capacity theory by modifying the bearing capacity factors and providing more accurate expressions for calculating soil resistance. Vesic's method is widely used in modern foundation engineering because it accounts for different soil conditions and foundation shapes.
These theoretical developments form the basis of modern geotechnical engineering and are still widely used for foundation design.

3.2 Experimental Studies on Soil Bearing Capacity

In addition to theoretical approaches, many researchers have conducted experimental studies to determine soil bearing capacity. Laboratory and field tests help engineers understand the behavior of soil under loading conditions.
One commonly used field test is the **Plate Load Test**, which measures the settlement of soil under a gradually increasing load. The results of this test help determine the safe bearing capacity of soil at a particular site.
Another widely used method is the **Standard Penetration Test (SPT)**, which measures the resistance of soil to penetration by a standard sampler. The SPT value provides information about soil strength and can be used to estimate bearing capacity indirectly.
Laboratory tests such as **triaxial compression tests**, **direct shear tests**, and **consolidation tests** are also used to determine soil strength parameters such as cohesion and internal friction angle. These parameters are essential for calculating soil bearing capacity using theoretical formulas.
Although experimental methods provide reliable results, they can be timeconsuming and expensive. Therefore, researchers have explored computational techniques to improve efficiency and reduce costs.

3.3 Numerical Methods in Geotechnical Engineering

With the advancement of computer technology, numerical methods have become widely used in geotechnical engineering. Numerical techniques allow engineers to simulate soil behavior under different loading conditions.

One commonly used numerical method is the **Finite Element Method (FEM)**. FEM divides the soil and structure into small elements and analyzes the stress and deformation behavior under applied loads. Software tools such as **PLAXIS, ABAQUS, and ANSYS** are commonly used for finite element analysis of soil–structure interaction problems.

Finite element models help engineers analyze complex problems such as foundation settlement, slope stability, and soil failure mechanisms. However, numerical methods require detailed input data and computational resources.

In recent years, data-driven approaches such as machine learning have been introduced to complement traditional numerical methods.

3.4 Application of Machine Learning in Civil Engineering

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions. Machine learning techniques are increasingly used in civil engineering to analyse large datasets and predict engineering parameters.

Machine learning algorithms can analyse relationships between multiple variables and produce accurate predictions without relying solely on theoretical formulas. These techniques are particularly useful when dealing with complex and nonlinear relationships between soil parameters.

Some commonly used machine learning algorithms in engineering applications include:

- Artificial Neural Networks (ANN)
- Support Vector Machines (SVM)
- Decision Trees
- Random Forest

These models have been successfully applied in areas such as structural health monitoring, traffic prediction, construction cost estimation, and geotechnical engineering.

3.5 Machine Learning for Soil Property Prediction

Several researchers have applied machine learning models to predict soil properties. These models use datasets containing soil parameters and corresponding engineering properties to train prediction models.

Artificial Neural Networks (ANN) have been widely used to predict soil properties such as compressibility, shear strength, and settlement. ANN models consist of interconnected nodes that simulate the functioning of the human brain. They are capable of learning complex nonlinear relationships between variables.

Studies have shown that ANN models can provide accurate predictions of soil properties when trained with sufficient data. However, neural networks sometimes require large datasets and careful tuning of model parameters.

Another machine learning method used in geotechnical engineering is **Support Vector Machines (SVM)**. SVM models are effective for classification and regression problems. They are particularly useful for small datasets and can handle nonlinear relationships using kernel functions.

In recent years, **Random Forest algorithms** have gained popularity due to their robustness and high prediction accuracy. Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction performance and reduce overfitting.

3.6 Random Forest in Engineering Applications

Random Forest is a machine learning algorithm developed by **Leo Breiman (2001)**. It works by creating multiple decision trees using random subsets of the training data. The predictions of these trees are then combined to produce the final output.

The main advantages of Random Forest include:

- High prediction accuracy
- Ability to handle large datasets
- Resistance to overfitting
- Capability to handle nonlinear relationships
- Automatic feature importance evaluation

Because of these advantages, Random Forest has been used in many engineering applications, including prediction of soil properties and structural behavior.

Researchers have used Random Forest models to estimate soil bearing capacity using soil parameters such as cohesion, friction angle, and unit weight. The results showed that Random Forest models can produce predictions that are comparable to traditional analytical methods.

3.7 Web-Based Engineering Applications

In addition to predictive modeling, modern engineering systems often include webbased interfaces to make the tools accessible to users. Web frameworks such as **Flask** are commonly used in Python to develop web applications.

Flask is a lightweight web framework that allows developers to create web-based systems for data input, prediction, and result visualization. Flask applications can integrate machine learning models and provide real-time predictions through a web interface.

By combining machine learning models with web applications, engineers can create user-friendly tools that allow users to input data and obtain predictions quickly.

CHAPTER 4

FUNDAMENTALS OF SOIL BEARING CAPACITY

In civil engineering, every structure such as buildings, bridges, dams, and roads is supported by soil through foundations. The soil must be strong enough to carry the loads coming from the structure. If the soil cannot support the load, the structure may settle excessively or even collapse.

The ability of soil to support structural loads is known as **soil bearing capacity**. It is one of the most important parameters in foundation engineering and plays a key role in safe and economical construction.

The concept of soil bearing capacity was first systematically explained by Karl Terzaghi, who developed the basic theory for calculating the ultimate load that soil can carry. His work formed the foundation of modern geotechnical engineering. Understanding soil bearing capacity helps engineers design safe foundations and select appropriate construction techniques.

4.1 Definition of Soil Bearing Capacity

Soil bearing capacity can be defined as **"The maximum load per unit area that soil can support without undergoing shear failure or excessive settlement."**

When a load is applied on soil through a foundation, the stress spreads into the ground. If the applied stress becomes greater than the soil strength, the soil may fail. Therefore, the load applied by the structure must always be less than the safe bearing capacity of the soil.

4.2 Importance of Bearing Capacity in Civil Engineering

Bearing capacity plays an essential role in foundation design. Its importance includes:

Structural Safety

If the bearing capacity is insufficient, the soil may fail and cause structural damage or collapse.

Economic Design

Accurate estimation prevents overdesign of foundations, which reduces construction cost.

Settlement Control

Proper bearing capacity analysis helps control excessive settlement of structures.

Foundation Selection

Knowledge of soil strength helps engineers choose the appropriate foundation type such as shallow or deep foundations.

4.3 Types of Bearing Capacity

1. Ultimate Bearing Capacity
2. Net Ultimate Bearing Capacity
3. Safe Bearing Capacity
4. Allowable Bearing Capacity

4.3.1 Ultimate Bearing Capacity

Ultimate bearing capacity is the **maximum pressure that soil can withstand before shear failure occurs**.

It is usually represented by:

$$q_u$$

When the applied load reaches this level, the soil experiences failure.

4.3.2 Net Ultimate Bearing Capacity

Net ultimate bearing capacity is the ultimate bearing capacity minus the overburden pressure at the foundation level.

$$q_{nu} = q_u - \gamma D_f$$

Where:

- γ= unit weight of soil
- D_f= depth of foundation

4.3.3 Safe Bearing Capacity

Safe bearing capacity is the maximum load that soil can safely support after applying a **factor of safety**.

$$q_{safe} = \frac{q_u}{FOS}$$

Where:

FOS = Factor of Safety (usually between 2.5 and 3).

4.3.4 Allowable Bearing Capacity

Allowable bearing capacity is the pressure that soil can safely support without causing excessive settlement.

It is generally the smaller value between:

- safe bearing capacity based on shear failure
- bearing capacity based on allowable settlement

4.4 Factors Affecting Soil Bearing Capacity

Many factors influence the bearing capacity of soil.

4.4.1 Soil Type

Different soils have different strength properties.

Examples:

Soil Type	**Bearing Capacity**
Clay	Low to moderate
Sand	Moderate to high
Gravel	High

Gravel soils generally have the highest bearing capacity.

4.4.2 Soil Cohesion

Cohesion is the internal bonding between soil particles.

Higher cohesion generally increases the bearing capacity of soil.

Cohesion is represented by:

$$c$$

4.4.3 Angle of Internal Friction

The angle of internal friction represents the resistance of soil particles to sliding over each other.

It is represented by:

$$\phi$$

Higher friction angle leads to higher bearing capacity.

4.4.4 Unit Weight of Soil

Unit weight affects the stress distribution in soil.

Higher soil density increases resistance against failure.

4.4.5 Depth of Foundation

Bearing capacity generally increases with increasing depth of foundation because of the confining pressure from surrounding soil.

4.4.6 Width of Footing

Larger foundation width distributes the load over a larger area, which increases the load carrying capacity.

4.4.7 Groundwater Level

Presence of groundwater reduces effective stress in soil, which decreases bearing capacity.

4.5 Terzaghi Bearing Capacity Theory

Terzaghi proposed the following equation for shallow foundations.

$$q_u = c' N_c + \gamma D N_q + 0.5 \gamma B N_\gamma$$

Where

- q_u = ultimate bearing capacity
- c' = cohesion
- γ = unit weight
- B = footing width
- D = foundation depth

4.6 Methods for Determining Bearing Capacity of Soil

In geotechnical engineering, determining the bearing capacity of soil is very important before constructing any structure. Engineers must know how much load the soil can safely support so that the foundation design will be stable and safe. Several field tests, laboratory tests, and theoretical methods are used to determine soil bearing capacity. These methods help engineers understand soil strength and behaviour under load. The fundamental theories used today were developed by the famous geotechnical engineer Karl Terzaghi, whose work established the scientific basis for foundation design.
The following are the main methods used to determine soil bearing capacity.

4.6.1 Plate Load Test

Introduction

The Plate Load Test is one of the most direct and reliable field methods used to determine the bearing capacity of soil. It measures the settlement of soil under a loaded plate placed at the foundation level.

Procedure

1. A test pit is excavated at the proposed foundation depth.
2. A rigid steel plate (usually 300 mm to 750 mm diameter) is placed at the bottom of the pit.
3. A hydraulic jack is used to apply load on the plate.
4. The load is applied in increments.
5. Settlement of the plate is measured using dial gauges.
6. The load–settlement curve is plotted.

Result Interpretation

The bearing capacity is determined from the load–settlement graph. The point where settlement increases rapidly indicates failure.

Advantages

- Provides direct measurement of soil strength
- Reliable for shallow foundations
- Simple test procedure

Limitations

- Expensive and time consuming
- Represents only a small area of soil
- Not suitable for deep foundations

4.6.2 Standard Penetration Test (SPT)

Introduction

The Standard Penetration Test is one of the most commonly used field tests for determining soil properties and bearing capacity. It is conducted during drilling operations.

Procedure

1. A borehole is drilled into the ground.
2. A split-spoon sampler is placed at the bottom of the borehole.
3. A hammer weighing 63.5 kg is dropped from a height of 750 mm.
4. The number of blows required to drive the sampler 300 mm into the soil is recorded.

The number of hammer blows is called the **SPT N-value**.

Interpretation

Higher N-values indicate stronger soil with higher bearing capacity.

Example:

Soil Type	**N-value**	**Bearing Capacity**
Soft clay	0–4	Very low
Medium sand	10–30	Moderate
Dense sand	30–50	High

Advantages

- Quick and economical
- Provides soil samples
- Widely used in foundation design

Limitations

- Results affected by operator skill
- Not accurate in very soft or gravelly soils

4.6.3 Cone Penetration Test (CPT)

Introduction

The Cone Penetration Test is a modern method used to determine soil strength and bearing capacity. It provides continuous soil data with depth.

Procedure

1. A cone-shaped probe is pushed into the ground at a constant rate.
2. Resistance offered by soil to the cone is measured.
3. The test records:

- cone resistance
- sleeve friction
- pore water pressure (in advanced CPT tests)

Interpretation

Higher cone resistance indicates stronger soil.

Advantages

- Continuous soil profile
- Accurate and reliable results
- Fast testing process

Limitations

- Requires special equipment
- Difficult to perform in gravelly soil

4.6.4 Laboratory Shear Tests

Soil samples collected from the field are tested in laboratories to determine shear strength parameters. These parameters help calculate soil bearing capacity.

4.6.5 Common laboratory tests include:

4.6.5.1 Direct Shear Test

In this test, a soil sample is placed in a shear box and subjected to horizontal shear force until failure occurs.

The test determines:

- soil cohesion
- angle of internal friction

These parameters are used in bearing capacity calculations.

4.6.5.2 Triaxial Compression Test

This is one of the most reliable laboratory tests for determining soil strength.

Procedure

1. A cylindrical soil sample is placed in a pressure chamber.
2. Confining pressure is applied around the sample.
3. Vertical load is applied until failure occurs.

The test helps determine:

- shear strength
- cohesion
- friction angle

These parameters are essential in geotechnical design.

CHAPTER 5

MACHINE LEARNING CONCEPTS

5.1 Definition

Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data and make predictions.

5.2 Introduction to Machine Learning

Machine Learning is a modern computer technology that allows computers to learn from data and make predictions without being directly programmed for every situation. It is a part of the field of Artificial Intelligence.

In simple words, machine learning teaches computers to find patterns in data and use those patterns to predict future results.

For example, if we give a computer many examples of soil properties and their corresponding bearing capacities, the computer can learn the relationship between them. After learning this relationship, it can predict the bearing capacity for new soil conditions.

In this project, machine learning is used to predict soil bearing capacity based on different soil parameters such as cohesion, friction angle, unit weight, foundation depth, and footing width.

5.3 Why Machine Learning is Used in Geotechnical Engineering

Soil behavior is complex and nonlinear. Traditional equations used in foundation engineering are based on simplified assumptions and may not represent real soil conditions accurately.

Machine learning models can analyze large datasets and identify complex relationships between soil parameters. This makes them very useful in modern Geotechnical Engineering.

Using algorithms from Machine Learning, engineers can develop models that provide faster and sometimes more accurate predictions of soil bearing capacity.

Benefits of using machine learning include:

5.4 Basic Working of Machine Learning

The working process of machine learning generally follows these steps:

Step 1: Data Collection

Data related to soil properties is collected from experiments, field tests, or published research.

Step 2: Data Preprocessing

Raw data is usually not ready for direct use. It must be cleaned and organized.

Step 3: Splitting the Dataset

The dataset is divided into two parts:

1. **Training Data** – used to train the machine learning model
2. **Testing Data** – used to evaluate the performance of the model Usually:

- 70–80% of data is used for training
- 20–30% of data is used for testing

Step 4: Model Training

In this stage, a machine learning algorithm learns patterns from the training data. The algorithm analyses how input parameters affect the output value (bearing capacity). After training, the model becomes capable of making predictions.

Step 5: Model Testing

The trained model is tested using the testing dataset.

The predicted values are compared with actual values to check the accuracy of the model.

Step 6: Prediction

After successful testing, the model can be used to predict soil bearing capacity for new soil conditions.

5.5 Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

5.5.1 Supervised Learning

In supervised learning, the model learns from labeled data. This means the input data and the correct output values are already known.

Example:

Input →Soil properties

Output → Bearing capacity

Since this project predicts a known output from input variables, it uses supervised learning.

5.5.2 Unsupervised Learning

In unsupervised learning, the model analyzes data without known output values. It tries to identify patterns or group similar data points.

Example:

Grouping soil types based on properties.

5.5.3 Reinforcement Learning

In reinforcement learning, the model learns by interacting with an environment and receiving rewards or penalties.

This method is mainly used in robotics and gaming applications.

5.6 Algorithms Used in This Project

❖ **Random Forest Model**

The **Random Forest model** is one of the most powerful **machine learning algorithms** used for prediction and classification. It belongs to the **ensemble learning method**, which means it combines multiple models to improve prediction accuracy. Random Forest works by creating many decision trees and combining their results to produce a final output.

Random Forest is widely used in engineering, finance, healthcare, and research because it provides **high accuracy, handles large datasets, and reduces overfitting**. Random Forest is implemented in Python using libraries like **Scikit-learn** with the programming language **Python**.

5.6.1 Concept of Random Forest

Random Forest is based on the concept of **multiple decision trees**.

Instead of relying on a single decision tree, the algorithm:

1. Creates many decision trees from random subsets of data
2. Each tree makes a prediction
3. The final result is the **average (for regression)** or **majority vote (for classification)**

Because many trees are used, the model becomes more **stable and accurate**.

5.6.2 Working Algorithm of Random Forest

Step-by-step process:

1. Input dataset with features and target values
2. Select random samples using bootstrap sampling
3. Build multiple decision trees
4. Each tree predicts output
5. Combine predictions using averaging

Final output = **best prediction**

5.6.3 Advantages of Random Forest

1. High prediction accuracy
2. Handles large datasets
3. Works with nonlinear relationships
4. Reduces overfitting
5. Works with missing data

5.6.4 Disadvantages of Random Forest

1. Requires more computation power
2. Slower than simple models
3. Difficult to interpret

Random Forest is widely used in engineering prediction problems.

5.6.5 These models can use inputs like:

- Cohesion (c)
- Angle of friction (φ)
- Unit weight (γ)
- Depth of footing (Df)
- Width of footing (B)

To Predict:

Ultimate Bearing Capacity

Safe Bearing Capacity (SBC).

CHAPTER 6

METHODOLOGY

The methodology of this project describes the systematic process used to develop a **machine learning-based system for predicting the bearing capacity of soil**. The project combines concepts from soil mechanics, data analysis, machine learning, and web application development. The main steps involved in the methodology include dataset generation, data preprocessing, model training, prediction, and development of a user interface for practical use.

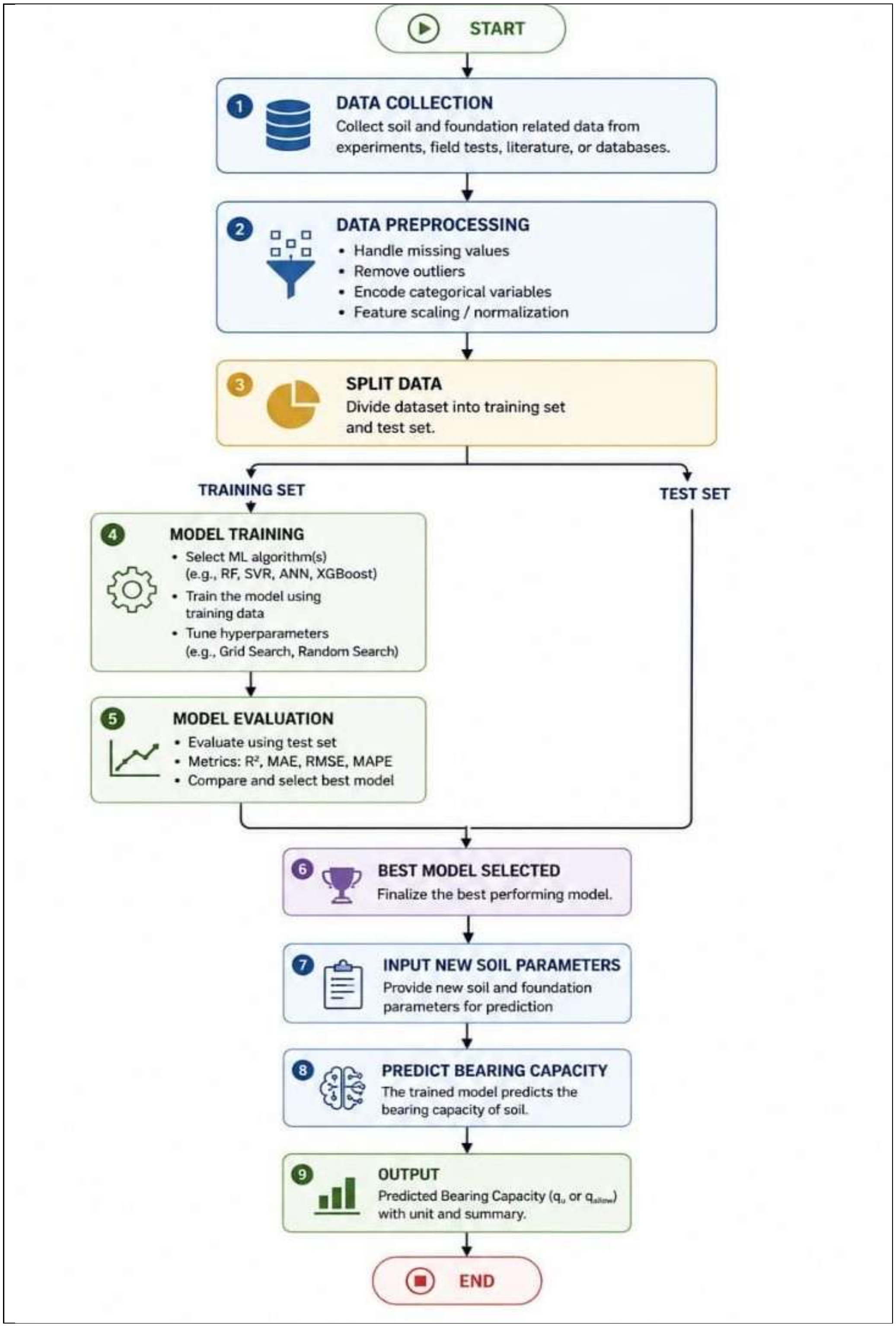
6.1 Understanding Soil Bearing Capacity Theory

The first step in the project is to study the basic concepts of **soil mechanics and bearing capacity theory**. In this project, **Terzaghi's bearing capacity equation** is used as the theoretical foundation for generating soil data. Terzaghi's equation calculates the **ultimate bearing capacity** of soil using important parameters such as cohesion (c), angle of internal friction (φ), unit weight of soil (γ), depth of foundation (Df), and width of footing (B). These parameters represent the physical properties of soil and foundation conditions.

The ultimate bearing capacity is calculated using Terzaghi's formula, and the **safe bearing capacity** is obtained by dividing the ultimate capacity by the **factor of safety (FOS)**. These values are used as the target outputs for the machine learning model.

6.2 System Design

6.2.1. System Architecture



6.2.2. Folder Structure

```
Predict_Soil_Bearing_Capacity/
│
├── dataset/
│    ├── soil_data.csv
│
├── models/
│    ├── soil_encoder.pkl
│    └── footing_encoder.pkl
│    ├── bearing_capacity_model.pkl
│
├── templates/
│    ├── register.html
│    ├── login.html
│    ├── predict.html
│    └── result.html
│
├── static/
│    ├── style.css
│    └── script.js
│
├── python/
│    └── database.py
│    ├── predict_and_store.py
│    └── train_model.py
│
└── app.py
```

Folder Description:

Folder	Function
dataset	Stores soil datasets
models	Trained machine learning model
templates	Fronted web pages
static	CSS & JS
app.py	Main application file

6.2.3 Dataset Collection

After understanding the theoretical model, the next step is to create a **soil dataset**. In this project, the dataset contains different combinations of soil parameters such as:

- ✓ Soil type
- ✓ Footing type
- ✓ Cohesion (c)
- ✓ Angle of internal friction (φ)
- ✓ Unit weight of soil (γ)
- ✓ Depth of foundation (Df)
- ✓ Width of footing (B)
- ✓ Factor of safety (FOS)

Using these parameters, the program calculates the **ultimate bearing capacity (qu)** and **safe bearing capacity (qs)** using machine learning model. Multiple data samples are generated to create a dataset suitable for machine learning model training.

Code for Dataset Loading:

```
df = pd.read_csv("dataset/soil_bearing_capacity_dataset.csv")

# Input & Output
X = df[
    [
        "Soil_Type",
        "Footing_Type",
        "Cohesion_c",
        "Angle_of_Friction_phi",
        "Unit_Weight_gamma",
        "Depth_of_Foundation_Df",
        "Width_B",
        "Factor_of_Safety"
    ]
]
y = df[
    [
```

```
        "Ultimate_Bearing_Capacity_qu",
        "Safe_Bearing_Capacity_qs"
    ]
]
```

Reference: train_model.py

6.2.4. Data Preprocessing

Before training the machine learning model, the dataset must be prepared properly. Some parameters in the dataset such as **soil type and footing type are categorical variables**. Machine learning algorithms require numerical inputs, so these categorical variables are converted into numerical values using **Label Encoding**.

After encoding the data, the dataset is divided into two parts:

Input features (X) – Soil parameters and foundation properties

Target variables (y) – Ultimate and safe bearing capacity values

The categorical parameters such as soil type and footing type were converted into numerical form using Label Encoding for machine learning processing.

Code snippet:

```
le_soil = LabelEncoder()
le_footing = LabelEncoder()
df["Soil_Type"] = le_soil.fit_transform(df["Soil_Type"])
df["Footing_Type"] = le_footing.fit_transform(df["Footing_Type"])
```

Reference: train_model.py

6.2.5.Train-Test Split

To evaluate the performance of the Machine Learning model, the dataset was divided into:

- Training Dataset
- Testing Dataset

The training dataset is used for model learning, while the testing dataset is used to evaluate prediction accuracy.

In this project:

- 80% data used for training
- 20% data used for testing

Code

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Reference: train_model.py

6.2.6. Model Training

In this project, the **Random Forest Regression algorithm** is used to train the prediction model. Random Forest is an ensemble machine learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

The dataset is divided into **training data and testing data** using the train–test split method. The training dataset is used to train the machine learning model, while the testing dataset is used to evaluate its performance.

The Random Forest Regressor algorithm was used for predicting ultimate and safe bearing capacity due to its high accuracy and ability to handle nonlinear relationships.

Code snippet:

```
model = RandomForestRegressor(
    n_estimators=200,
    random_state=42
)
model.fit(X_train, y_train)
```

Reference: train_model.py

6.2.7. Model Evaluation

After training the model, its performance is evaluated using the **R^2 (R-squared) score**. The R^2 score measures how well the model predicts the target values compared to actual values. A higher R^2 score indicates better prediction accuracy.

Formula of R^2 Score

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Where:

- y_i= Actual value
- $\hat{y}_i$= Predicted value
- $\bar{y}$= Mean value

Code snippet:

```
accuracy = r2_score(y_test, y_pred)
print("Model Accuracy (R² Score):", accuracy)
```

Reference: train_model.py

6.2.8. Model Saving

Once the model is trained successfully, it is saved using the **Joblib library**. Saving the model allows it to be reused later without retraining. The label encoders used for categorical variables are also saved to ensure consistency during prediction.

Code snippet:

```
joblib.dump(model, "bearing_capacity_model.pkl")
joblib.dump(le_soil, "soil_encoder.pkl")
joblib.dump(le_footing, "footing_encoder.pkl")


print("Model and encoders saved successfully!")
```

Reference: train_model.py

6.3. Prediction System Development

To make the system practical and user-friendly, a **Flask-based web application** is developed. The web interface allows users to input soil parameters such as soil type, cohesion, friction angle, foundation depth, and footing width. The application sends this data to the trained machine learning model, which predicts the **ultimate and safe bearing capacity** instantly.

6.3.1. Flask Application Configuration

Purpose

This module initializes the Flask web application and imports all required libraries for database management, machine learning prediction, session handling, and data processing.

Code

```
from flask import Flask, render_template, request, redirect, url_for, session
import sqlite3
import pandas as pd
import joblib
import os

app = Flask(__name__)
app.secret_key = "soil_bearing_capacity_secret_key"

DB_NAME = "bearing_capacity.db"
```

Reference: app.py

Explanation

- Flask is used to create the web application.
- sqlite3 manages the database.
- pandas handles input data formatting.
- joblib loads trained machine learning models.
- secret_key is used for user session security.
- DB_NAME stores the database filename.

6.3.2. Loading Machine Learning Model

Purpose

This section loads the trained machine learning model and categorical encoders used for prediction.

Code

```
model = joblib.load("bearing_capacity_model.pkl")
soil_encoder = joblib.load("soil_encoder.pkl")
footing_encoder = joblib.load("footing_encoder.pkl")
```

Reference: app.py

Explanation

- bearing_capacity_model.pkl stores the trained ML model.
- Encoders convert categorical values into numerical format.
- These files are loaded before prediction starts.

6.3.3. Database Connection Module

Purpose

This module establishes connection with the SQLite database.

Code

```
def get_db():
    conn = sqlite3.connect(DB_NAME)
    conn.row_factory = sqlite3.Row
    return conn
```

Reference: app.py

Explanation

- Creates connection with the database.
- row_factory enables accessing columns by name.
- Returns database connection object.

6.3.4. Database Initialization

Purpose

This module creates tables for storing user information and prediction records.

Code

```
def init_db():
    conn = get_db()
    cur = conn.cursor()

    cur.execute("""
    CREATE TABLE IF NOT EXISTS users (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        username TEXT UNIQUE,
        password TEXT
    )
    """)
```

Reference: app.py

Explanation

- Creates user's table.
- AUTOINCREMENT automatically generates user ID.
- UNIQUE prevents duplicate usernames.

6.3.5. Prediction Table Creation

Code

```
cur.execute("""
CREATE TABLE IF NOT EXISTS predictions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    username TEXT,
    soil TEXT,
    footing TEXT,
    cohesion REAL,
```

```
    phi REAL,
    gamma REAL,
    depth REAL,
    width REAL,
    fos REAL,
    ultimate REAL,
    safe REAL
)
""")
```

Reference: app.py

Explanation

This table stores:

Soil type, Footing type, Cohesion value, Angle of friction, Unit weight of soil, Foundation dimensions, Factor of safety, and predicted ultimate and safe bearing capacities.

6.3.6. User Registration Module

Purpose

Allows new users to create an account.

Code

```
@app.route("/register", methods=["GET", "POST"])
def register():
```

Reference: app.py

Explanation

- Verifies username and password.
- Creates user session after successful login.
- Redirects user to prediction page.

6.3.7. Prediction Module

Purpose

This module performs soil bearing capacity prediction using the trained machine learning model.

Code

```
ultimate, safe = model.predict(X)[0]
ultimate = round(ultimate, 2)
safe = round(safe, 2)
```

Reference: app.py

Explanation

- Input values are converted into DataFrame format.
- Encoders transform categorical variables.
- ML model predicts:
    - Ultimate Bearing Capacity
    - Safe Bearing Capacity
- Results are rounded to two decimal places.

6.3.8. Saving Prediction Results

Code

```
cur.execute("""
INSERT INTO predictions (
    username, soil, footing, cohesion, phi,
    gamma, depth, width, fos, ultimate, safe
)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
""")
```

Reference: app.py

Explanation

- Stores prediction results in database.
- Maintains history of all user predictions.

6.3.9. Logout Module

Purpose

Ends user session securely.

Code

```
@app.route("/logout")
def logout():
    session.clear()
    return redirect(url_for("login"))
```

Reference: app.py

Explanation

- Clears all session data.
- Redirects user back to login page.

6.3.10. Running the Application

Code

```
if __name__ == "__main__":
    init_db()
    app.run(debug=True)
```

Reference: app.py

Explanation

- Initializes database before application starts.
- Runs Flask development server.
- debug=True helps identify errors during development.

6.4. Database Integration

The project also includes a **SQLite database** to store prediction results. Whenever a user performs a prediction, the input parameters and predicted results are saved in the database. This helps in maintaining a record of previous predictions for analysis and reference.

SQLite database was used for storing:

- User registration details
- Login information
- Prediction records

The database helps maintain prediction history and improve data management.

6.4.1. Database Connection

Code:

```
import sqlite3
conn = sqlite3.connect("bearing_capacity.db")
cursor = conn.cursor()
```

Reference: database.py

6.4.2. Table Creation

Code:

```
cursor.execute("""
CREATE TABLE IF NOT EXISTS predictions
 (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    soil_type TEXT,
    footing_type TEXT,
    cohesion REAL,
    phi REAL,
    gamma REAL,
    depth REAL,
    width REAL,
    fos REAL,
    ultimate_capacity REAL,
```

```
    safe_capacity REAL
)
""")
```
Reference: app.py

6.5. Conclusion

Finally, the predicted values of **ultimate bearing capacity and safe bearing capacity** are displayed on the web interface. This allows users to quickly analyze soil conditions and make decisions related to foundation design.

Overall, the methodology of this project integrates **soil mechanics theory, machine learning techniques, and web development tools** to create a complete system for predicting soil bearing capacity efficiently. **R^2 value is 0.98 (98 %) which indicates higher accuracy of trained data with tested data.**

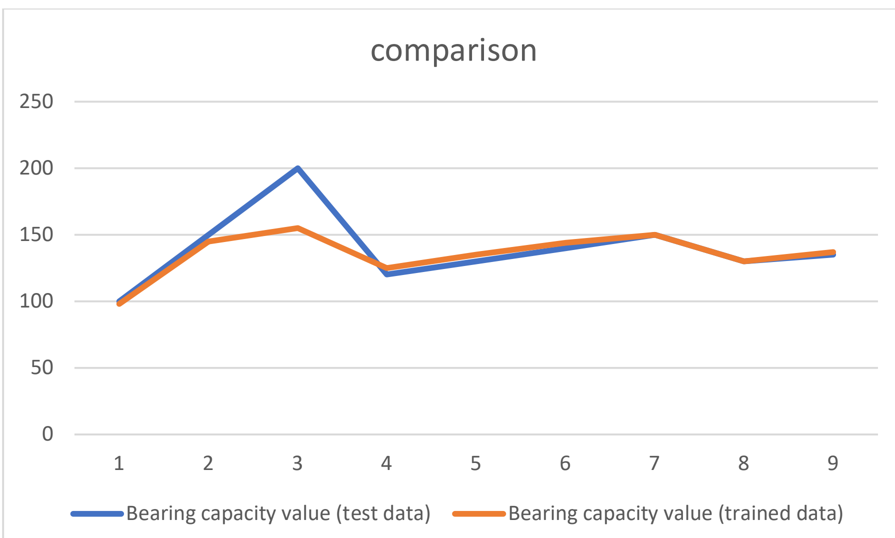
The project highlights the integration of **civil engineering domain knowledge with computer applications**, demonstrating the potential of ML for improving geotechnical design efficiency.

CHAPTER 7

RESULT & CONCLUSION

Interface

s.no.	Bearing capacity value (test data)	Bearing capacity value (trained data)
1	100	98
2	150	145
3	200	155
4	120	125
5	130	135
6	140	144
7	150	150
8	130	130
9	135	137



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Soil_Type	Footing_Typ	Cohesion_	Angle_of_	Unit_Weig	Depth_of_	Width_B	Factor_of_	Nc	Nq	N_gamma	Ultimate_I	Safe_Beari
Alluvial	Combined	18.67	20	17.85	2.07	1.48	3	14.83	6.4	5.39	584.55	194.85
Clayey Lo	Isolated(Circ	23.03	20	18.86	1.97	1.09	3	14.83	6.4	5.39	634.72	211.57
Alluvial	Combined	20.1	20	17.86	0.97	1.12	3	14.83	6.4	5.39	462.87	154.29
Silty Clay I	Combined	24.73	25	18.56	0.99	1.45	2.5	20.72	10.66	10.88	854.68	341.87
Silty Clay I	Combined	12.56	30	17.97	1.09	1.23	3	30.14	18.4	22.4	986.52	328.84
Silty Clay I	Isolated (Sq	20.94	30	17.14	2.02	1.78	3	30.14	18.4	22.4	1609.89	536.63
Clayey Lo	Isolated(Rec	13.56	30	18.35	1.23	2.8	3	30.14	18.4	22.4	1399.45	466.48
Clayey Lo	Combined	23.64	30	17.56	1.42	1.94	3	30.14	18.4	22.4	1552.86	517.62
Silty Clay I	Combined	15.81	20	17.49	2.31	1.55	2.5	14.83	6.4	5.39	566.09	226.44
Alluvial	Isolated(Rec	28.94	20	17.02	2.32	2.48	3	14.83	6.4	5.39	795.65	265.22
Clayey Lo	Isolated (Sq	20.95	25	17.95	0.91	2.97	2.5	20.72	10.66	10.88	898.22	359.29
Alluvial	Isolated (Sq	15.64	30	18.44	2.47	2.65	3	30.14	18.4	22.4	1856.75	618.92
Clayey Lo	Isolated (Sq	29.41	20	18.04	1.55	1.66	2.5	14.83	6.4	5.39	695.81	278.32
Clayey Lo	Isolated (Sq	28.46	30	17.97	2.44	2.63	2.5	30.14	18.4	22.4	2193.89	877.56
Clayey Lo	Combined	29.43	30	18.46	2.16	2.51	3	30.14	18.4	22.4	2139.64	713.21
Alluvial	Strip	10.98	20	17.6	1.79	1.47	2.5	14.83	6.4	5.39	434.18	173.67
Clayey Lo	Combined	21.29	30	17.88	1.75	1.24	3	30.14	18.4	22.4	1465.73	488.58
Silty Clay I	Isolated (Sq	10.33	30	17.36	1.14	2.78	3	30.14	18.4	22.4	1216.01	405.34
Silty Clay I	Isolated(Rec	11.44	25	18.98	1.09	1.47	3	20.72	10.66	10.88	609.35	203.12
Alluvial	Combined	24.84	25	18.35	1.2	2.61	3	20.72	10.66	10.88	1009.96	336.65
Clayey Lo	Isolated(Rec	15.65	25	18.26	2.24	2.16	3	20.72	10.66	10.88	974.85	324.95
Clayey Lo	Isolated(Circ	19.91	20	17.32	1.58	1.16	3	14.83	6.4	5.39	524.55	174.85
Clayey Lo	Isolated(Rec	17.73	20	17.07	1.91	1.13	2.5	14.83	6.4	5.39	523.58	209.43
Alluvial	Strip	26.57	30	17.15	2.17	2.3	3	30.14	18.4	22.4	1927.37	642.46
Clayey Lo	Strip	19.16	20	17.28	1.75	1.39	3	14.83	6.4	5.39	542.41	180.8
Silty Clay I	Isolated(Circ	20.82	25	17.05	1.83	1.55	3	20.72	10.66	10.88	907.76	302.59
Alluvial	Isolated(Circ	17.14	20	18.47	1.23	1.03	3	14.83	6.4	5.39	450.85	150.28
Silty Clay I	Isolated(Rec	11.3	25	18.6	1.33	1.8	2.5	20.72	10.66	10.88	679.97	271.99
Alluvial	Combined	22.59	20	17.1	2.48	2.44	3	14.83	6.4	5.39	718.87	239.62
Silty Clay I	Combined	13.11	25	17.95	2.48	2.97	2.5	20.72	10.66	10.88	1036.19	414.48
Silty Clay I	Isolated (Sq	15.96	30	17.63	1.17	1.2	3	30.14	18.4	22.4	1097.52	365.84
Clayey Lo	Isolated (Sq	28.58	20	17.82	2.1	2.36	3	14.83	6.4	5.39	776.68	258.89
Clayey Lo	Isolated (Sq	21.17	20	18.72	1.84	1.21	3	14.83	6.4	5.39	595.44	198.48
Silty Clay I	Isolated (Sq	15.95	25	18.03	2.19	1.51	3	20.72	10.66	10.88	899.51	299.84
Clayey Lo	Isolated(Circ	11.1	25	17.39	2.16	2.83	3	20.72	10.66	10.88	898.13	299.38
Silty Clay I	Isolated(Circ	22.15	20	17.93	1.75	2.43	3	14.83	6.4	5.39	646.72	215.57
Silty Clay I	Isolated (Sq	12.05	30	18.06	1.2	2.74	3	30.14	18.4	22.4	1316.18	438.73
Clayey Lo	Strip	11.12	30	17.08	2.13	2.85	3	30.14	18.4	22.4	1549.75	516.58
Alluvial	Isolated (Sq	27.62	25	17.37	2	1.47	2.5	20.72	10.66	10.88	1081.52	432.61
Silty Clay I	Combined	29.55	30	18.14	0.98	1.79	2.5	30.14	18.4	22.4	1581.41	632.56
Silty Clay I	Strip	14.51	20	17.02	0.99	2.8	3	14.83	6.4	5.39	451.45	150.48
Clayey Lo	Isolated(Rec	10.4	25	17.54	1.33	2.76	3	20.72	10.66	10.88	727.52	242.51
Silty Clay I	Isolated(Circ	22.39	20	18.19	1.85	1.97	3	14.83	6.4	5.39	643.99	214.66
Silty Clay I	Isolated (Sq	13.39	30	18.06	1.73	1.57	3	30.14	18.4	22.4	1296.03	432.01
Alluvial	Isolated (Sq	27.96	20	18.37	1.5	2.19	2.5	14.83	6.4	5.39	699.42	279.77
Silty Clay I	Strip	15.88	25	18.34	1.32	2.65	2.5	20.72	10.66	10.88	851.49	340.6
Silty Clay I	Combined	28.13	20	18.73	2.17	1.2	3	14.83	6.4	5.39	737.86	245.95
Clayey Lo	Strip	18.52	30	17.03	2.47	2.72	2.5	30.14	18.4	22.4	1850.97	740.39
Silty Clay I	Isolated (Sq	10.98	30	17.94	2.16	1.68	2.5	30.14	18.4	22.4	1381.5	552.6
Silty Clay I	Isolated (Sq	11.41	30	17.34	2.15	1.74	2.5	30.14	18.4	22.4	1367.79	547.12
Silty Clay I	Combined	14.93	25	17.09	1.22	1.32	3	20.72	10.66	10.88	654.33	218.11
Clayey Lo	Isolated (Sq	23.69	30	18.13	1.16	1.79	2.5	30.14	18.4	22.4	1464.45	585.78
Alluvial	Isolated (Sq	27.84	25	17.86	0.86	2.55	3	20.72	10.66	10.88	988.33	329.44
Silty Clay I	Isolated (Sq	22.95	25	17.74	1.58	2.94	2.5	20.72	10.66	10.88	1058.04	423.22
Alluvial	Isolated (Sq	20.31	20	18.13	1.11	1.55	3	14.83	6.4	5.39	505.73	168.58
Clayey Lo	Combined	22.37	25	17.92	1.84	1.88	2.5	20.72	10.66	10.88	998.27	399.31
Alluvial	Isolated (Sq	19.4	30	17.06	1.66	2.37	2.5	30.14	18.4	22.4	1558.64	623.46
Silty Clay I	Isolated(Rec	28.14	30	17.08	2.09	2.17	3	30.14	18.4	22.4	1920.08	640.03
Alluvial	Combined	15.12	25	17.59	2.31	1.13	2.5	20.72	10.66	10.88	854.56	341.82
Silty Clay I	Strip	17.33	25	18.37	1.86	2.47	3	20.72	10.66	10.88	970.14	323.38
Clayey Lo	Strip	16.39	30	17.57	2.49	2.3	3	30.14	18.4	22.4	1751.58	583.86
Alluvial	Isolated(Rec	28.93	25	17.35	1.4	2.73	2.5	20.72	10.66	10.88	1116.03	446.41
Clayey Lo	Isolated(Rec	22.99	25	18.36	1.63	1.32	3	20.72	10.66	10.88	927.21	309.07
Silty Clay I	Strip	13.12	20	17.77	2.16	2.86	3	14.83	6.4	5.39	577.19	192.4
Alluvial	Isolated (Sq	23.33	25	18.16	1.61	2.46	3	20.72	10.66	10.88	1038.09	346.03

Clayey Lo	Strip	16.11	20	17.35	1.1	2.92	2.5	14.83	6.4	5.39	497.59	199.04
Silty Clay I	Isolated (Sq	28.64	30	17.28	2.42	1.02	2.5	30.14	18.4	22.4	1830.06	732.02
Alluvial	Isolated (Sq	26.13	30	18.69	1.33	1.21	3	30.14	18.4	22.4	1498.23	499.41
Silty Clay I	Isolated(Circ	17.21	30	18.43	1.06	1.1	2.5	30.14	18.4	22.4	1105.23	442.09
Clayey Lo	Strip	15.03	20	18.83	2.39	1.93	2.5	14.83	6.4	5.39	608.86	243.54
Clayey Lo	Combined	18.12	30	18.12	2.24	2.71	3	30.14	18.4	22.4	1842.95	614.32
Alluvial	Strip	11.46	20	18.13	1.85	2.61	3	14.83	6.4	5.39	512.14	170.71
Clayey Lo	Strip	16.57	20	18.93	1.57	2.59	3	14.83	6.4	5.39	568.07	189.36
Clayey Lo	Strip	19.35	30	17.59	1.83	3	3	30.14	18.4	22.4	1766.52	588.84
Alluvial	Isolated(Rec	23.18	30	18.32	1.52	1.97	2.5	30.14	18.4	22.4	1615.23	646.09
Silty Clay I	Combined	28.49	20	17.57	0.92	2.92	3	14.83	6.4	5.39	664.22	221.41
Clayey Lo	Isolated (Sq	25.05	20	17.34	1.35	2.29	3	14.83	6.4	5.39	628.32	209.44
Clayey Lo	Isolated (Sq	10.76	25	18.27	2.02	2.17	3	20.72	10.66	10.88	832.03	277.34
Silty Clay I	Strip	20.47	20	18.72	1.78	1.62	2.5	14.83	6.4	5.39	598.56	239.42
Clayey Lo	Strip	28.24	20	18.13	2.08	1.73	2.5	14.83	6.4	5.39	744.67	297.87
Alluvial	Isolated(Rec	28.04	30	17.8	1.8	1.58	2.5	30.14	18.4	22.4	1749.65	699.86
Alluvial	Isolated(Rec	26.78	30	18.75	1.12	2.56	3	30.14	18.4	22.4	1731.15	577.05
Clayey Lo	Strip	20.38	30	18.97	1.8	2.14	3	30.14	18.4	22.4	1697.21	565.74
Clayey Lo	Strip	27.8	25	18.95	1.47	1.89	2.5	20.72	10.66	10.88	1067.8	427.12
Alluvial	Isolated (Sq	18.87	25	18.04	1.49	1.33	3	20.72	10.66	10.88	808.05	269.35
Clayey Lo	Isolated(Rec	26.56	20	18.03	1.09	1.36	2.5	14.83	6.4	5.39	585.75	234.3
Alluvial	Isolated (Sq	20.09	30	17.27	2.09	2.21	3	30.14	18.4	22.4	1697.11	565.7
Alluvial	Isolated(Rec	22.59	25	18.55	1.08	2.49	3	20.72	10.66	10.88	932.9	310.97
Alluvial	Isolated (Sq	29.81	25	18.1	1.72	1.02	2.5	20.72	10.66	10.88	1049.96	419.98
Alluvial	Strip	14.68	30	17.31	1.36	2.26	3	30.14	18.4	22.4	1313.77	437.92
Silty Clay I	Isolated(Circ	17.01	30	17.92	2.16	1.51	3	30.14	18.4	22.4	1527.96	509.32
Clayey Lo	Isolated(Circ	27.15	20	18.35	1.5	1.98	2.5	14.83	6.4	5.39	676.71	270.68
Silty Clay I	Strip	26.04	25	17.05	2.42	1.23	3	20.72	10.66	10.88	1093.48	364.49
Silty Clay I	Combined	23.56	25	17.48	0.91	2.11	3	20.72	10.66	10.88	858.37	286.12
Silty Clay I	Isolated(Rec	25.34	25	18.21	1.63	2.9	3	20.72	10.66	10.88	1128.74	376.25
Clayey Lo	Combined	13.18	30	17.81	1.28	2.69	2.5	30.14	18.4	22.4	1353.29	541.32
Alluvial	Strip	17.86	30	18.72	1.81	2.08	2.5	30.14	18.4	22.4	1597.85	639.14
Silty Clay I	Isolated(Rec	29.8	30	17.34	1.38	1	2.5	30.14	18.4	22.4	1532.68	613.07
Alluvial	Isolated(Circ	16.86	30	17.42	1.79	2.94	3	30.14	18.4	22.4	1655.51	551.84
Alluvial	Combined	22.87	30	18.67	2.47	1.74	2.5	30.14	18.4	22.4	1901.66	760.66
Clayey Lo	Strip	14.78	30	18.79	1.97	1.7	2.5	30.14	18.4	22.4	1484.33	593.73
Silty Clay I	Isolated (Sq	11.24	30	17.02	2.21	2.9	2.5	30.14	18.4	22.4	1583.68	633.47
Alluvial	Isolated(Circ	13.51	20	18.6	0.8	2.46	3	14.83	6.4	5.39	418.9	139.63
Silty Clay I	Isolated(Circ	13.46	30	18.5	2.4	2.17	3	30.14	18.4	22.4	1672.27	557.42
Silty Clay I	Combined	29.9	20	18.93	0.89	1.28	3	14.83	6.4	5.39	616.54	205.51
Silty Clay I	Isolated(Rec	12.42	30	18.82	1.56	1.85	2.5	30.14	18.4	22.4	1304.5	521.8
Clayey Lo	Strip	19.88	30	17.91	1.17	2.08	3	30.14	18.4	22.4	1401.98	467.33
Clayey Lo	Combined	26.61	25	17.38	1.75	1.41	2.5	20.72	10.66	10.88	1008.89	403.56
Alluvial	Isolated (Sq	14.06	30	18.33	1.38	1.86	2.5	30.14	18.4	22.4	1271.05	508.42
Alluvial	Combined	18.44	20	18.94	0.91	1.16	3	14.83	6.4	5.39	442.98	147.66
Silty Clay I	Strip	28.26	20	17.21	1.47	1.01	3	14.83	6.4	5.39	627.85	209.28
Alluvial	Strip	17.4	30	18.16	1.59	2.28	2.5	30.14	18.4	22.4	1519.46	607.78
Clayey Lo	Combined	28.61	25	18.62	1.59	2.98	2.5	20.72	10.66	10.88	1210.25	484.1
Silty Clay I	Isolated (Sq	10.89	20	18.92	1.93	1.04	2.5	14.83	6.4	5.39	448.23	179.29
Silty Clay I	Strip	24.74	20	18.66	1.84	1.83	2.5	14.83	6.4	5.39	678.66	271.46
Clayey Lo	Isolated (Sq	20.43	20	18.44	2.14	2.49	3	14.83	6.4	5.39	679.27	226.42
Silty Clay I	Isolated(Circ	18.06	25	17.55	1.27	2.05	2.5	20.72	10.66	10.88	807.52	323.01
Alluvial	Isolated (Sq	15.2	30	18.14	2.08	2.22	2.5	30.14	18.4	22.4	1603.42	641.37
Silty Clay I	Combined	15.02	25	17.05	2.46	2.57	3	20.72	10.66	10.88	996.7	332.23
Alluvial	Isolated (Sq	29.89	25	17.17	2.03	2.12	2.5	20.72	10.66	10.88	1188.89	475.56
Silty Clay I	Isolated(Rec	22.44	20	17.49	1.2	1.55	2.5	14.83	6.4	5.39	540.17	216.07
Clayey Lo	Isolated(Circ	15.37	25	17.08	0.83	2.58	3	20.72	10.66	10.88	709.31	236.44
Clayey Lo	Isolated(Rec	20.39	25	18.38	1.86	2.55	2.5	20.72	10.66	10.88	1041.88	416.75
Alluvial	Combined	13.63	20	18.29	1.91	2.49	2.5	14.83	6.4	5.39	548.45	219.38
Silty Clay I	Isolated(Circ	10.5	20	18.95	1.02	1.27	2.5	14.83	6.4	5.39	344.28	137.71
Alluvial	Isolated(Rec	25.21	30	17.55	2.05	1.43	3	30.14	18.4	22.4	1702.9	567.63
Alluvial	Strip	25.29	25	17.47	1.82	2.67	3	20.72	10.66	10.88	1116.7	372.23
Silty Clay I	Isolated(Circ	23.6	20	18.79	1.39	1.35	3	14.83	6.4	5.39	585.51	195.17
Silty Clay I	Isolated (Sq	18.15	30	18.26	2.43	2.45	3	30.14	18.4	22.4	1864.54	621.51
Alluvial	Isolated(Circ	25.77	25	18.34	1.13	2.11	2.5	20.72	10.66	10.88	965.39	386.16
Silty Clay I	Isolated (Sq	23.14	25	18.89	2.35	1.47	2.5	20.72	10.66	10.88	1103.73	441.49